The Case of Normandie Hydroliennes: A Cost-Benefit Analysis

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Abstract

This study conducts a cost–benefit analysis (CBA) of the NH1 offshore tidal energy project, which was developed as part of the Normandie Hydroliennes initiative. It aims to evaluates economic and social efficiency of the NH1 project by comparing costs and benefits over a 25-year period. Benefits include revenues from electricity generation and reduced CO_2 emissions, while costs include capital investment and operating expenditure. To avoid double-counting, it incorporates labour effects through a shadow-wage adjustment to reflect the social opportunity cost of employment, rather than as a direct separated benefit. The findings serve as a basis for evaluating, ex-ante, the project's contribution to France's low-carbon transition.

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1 Introduction

The *Normandie Hydroliennes* (NH1) project represents one of the first pilot tidal turbine farms in France. Located in the Raz Blanchard, one of the most promising sites worldwide for harnessing tidal stream energy, this project aims to install four next-generation turbines to generate local renewable electricity. The initiative, financed by the European Union (EU Innovation Fund grant €31.3M), is part of the broader European effort to diversify the energy mix and accelerate the transition towards low-carbon sources[1]. Tidal energy has particular advantages compared to other marine renewable technologies. Unlike wind or solar power, it relies on a predictable and regular resource, directly linked to tidal cycles. Several pilot projects, such as *MeyGen* in Scotland, has already demonstrated the technical feasibility and growing competitiveness of this technology [2]. However, NH1 stands out due to its location in a zone with exceptionally strong currents, allowing significant production with a limited number of turbines. In addition, tidal energy installations generally have a smaller visual footprint than offshore wind farms and do not produce intermittent fluctuations as severe as solar, offering a more stable contribution to the grid. Beyond electricity production, NH1 contributes to the emergence of a local industrial ecosystem. By demonstrating operational reliability, cost efficiency, and environmental compatibility, NH1 provides valuable lessons for scaling up tidal energy and integrating it effectively into the national energy strategy.

The objective of this case study is to conduct a Cost-Benefit Analysis (CBA) of the NH1 project in order to assess its economic efficiency and social desirability. By definition, according to OECD (2018) a Cost-Benefit Analysis provides a model of rationality and forces the decision makers to take into account the beneficiaries and losers for the implementation of a new project. In addition to the classical financial benefits and costs, the benefits are also defined as the increase in the well-being of an individual and the costs as a reduction of the well-being. This analysis follows the standard steps of a Cost-Benefit Analysis (CBA). First, we identify the underlying problem that justifies the implementation of a new project. Second, the policy scenario identification. Third, we define the scope of the analysis, defining the standing and the time horizon. Fourth, we list all the potential consequences arising from the implementation of the project. Fifth, the considered consequences are quantified using appropriate indicators, and sixth, they are monetized. Finally, all costs and benefits are aggregated and discounted in order to determine whether the project is socially desirable.

2 Step 1 : Problem Identification

The NH1 project aims to address France's need to increase renewable energy production and reduce carbon emissions. Indeed, even though nuclear energy is also a non-carbon intensive source of energy, according to OECD Nuclear Energy Agency and International Energy Agency

[1] "Le projet NH1", Normandie Hydroliennes
[2] "MeyGen Tidal Energy Project", Tethys

(2021), the average time from project initiation to operation for new reactors typically exceeds ten years. In contrast, pilot-scale tidal projects such as NH1 can progress from permitting to commissioning within five to six years. Thus, the urgence of reducing CO_2 emissions clearly call for a mix with alternative, faster to implement, sources of power. Furthermore, without the project, the region would remain reliant on a mix dominated by nuclear energy and on residual carbon-intensive generation. The NH1 project thus helps diversify the regional mix with a flexible, locally produced renewable source.

3 Step 2 : Policy Scenario Identification

Winner of the 2015 "Programmes d'Investissement d'Avenir" (PIA) program, the NH1 project aims to install four tidal turbines with a total capacity of 12 MW at Raz Blanchard in France. The farm is expected to produce around 33.9 GWh per year, supplying electricity to approximately 15,000 local inhabitants. Around 80% of the project's value is generated by French suppliers, creating about 400 direct and indirect jobs across the country. The total investment cost is estimated at €90 million, including design, construction, and installation which is €22.5 million per turbine. The turbines are expected to operate for around 25 years. However, since tidal technology remains relatively new, this estimate is based on empirical research and data from comparable previous projects.

4 Step 3 : Scope of the analysis - affected population and time horizon

This analysis will evaluate the costs and benefits over the expected lifetime of the turbines, which is approximately 25 years. Additionally, as the project is located in France, it is analysed within a French context. This considers both the local impact on a nearby village of 15,000 people, and the broader national effects.

5 Step 4 : List the consequences

The NH1 project will produce 33.9 GWh per year of low-carbon electricity, avoiding CO_2 emissions compared to fossil generation, reducing dependence on external fuels and exposure to price volatility, and diversifying the French low-carbon electricity mix. According to MacGillivray et al. (2014), marine renewable energy provides technology learning and demonstration value for future developments. The project also involves local industrial participation (80% French supply chain) and creates around 400 direct and indirect jobs in France.

Concerning the costs of the project, it includes capital expenditures for turbines, foundations, subsea cabling, and onshore integration; pre-deployment R&D. Peplinski et al. (2021) found that environmental permitting and compliance costs for marine energy projects includ-

ing pre-deployment surveys, environmental impact studies, and licensing represent a significant portion of total project expenses around 15%.

In addition, a common practice for the economic valuation of externalities related to a project is to use choice experiments (CE). However, while it has been done extensively for many environmental projects, externalities related to tidal turbine has not been explored as much yet (Vazquez and Iglesias, 2015). Hence, we can try to get results from similar projects by using offshore wind turbines (OWT) as a proxy. The discret choice experiments (DCE) method involve to build different scenarios where for each one, we modify format or level of some characteristics (Hanley et al., 2001). Then, surveying preferences between each scenario allow to elicit willingness-to-pay for different variations of those characteristics.

Therefore, we need to choose which characteristics are relevant. For instance, based on Mattmann et al. (2016), who conduct the first quantitative meta-analysis for non-market valuation literature on the external effects associated with wind power production, externalities considered for wind-turbines are related to climate change (*e.g.* CO_2 reduction), landscape (*e.g.* some people find unpleasant to see offshore or onshore wind turbines), or biodiversity (*e.g.* noise during construction phase can disrupt life on the area for up to 10 years). However, although tidal turbines and OWT share some externalities, tidal turbines have many advantages from being underwater, which makes some risks unlikely enough for us to consider them insignificant. Consequently, we ignore landscape externalities. NH1 project communicates that some in-depth studies have been made to assess the impact of the tidal turbines on the marine environment. Those studies conclude that, although there can be a small displacement at the installation, species return thanks to an "artificial reef" effect. In addition, they found that the sound pressure levels of tidal turbines are lower than the disturbance thresholds.[3]

Furthermore, Joalland and Mahieu (2023) found that for OWT, results are strongly sensitive to the impact on the fishing sector employment and the origin of sea food. These effects are caused by displacing the fishermen from traditional fishing areas. As mention previously, we assume tidal turbines to not be concerned by these externalities. However, we can use this result to justify the importance of employment in people's preferences. To avoid double counting, we will proceed by using a shadow wage price to transform the Capital Expenditures (CAPEX) and Operating Expenditures (OPEX) into Social Net CAPEX and Social Net OPEX.

6 Step 5 : Quantify the consequences

6.1 Benefits

6.1.1 Revenue from Electricity Generated

To quantify the revenue from the NH1 project, a key variable to estimate is the price at which Normandie Hydroliennes will sell the electricity generated by tidal turbines. Since there is no

[3]Those studies may me subject to conflict of interests, but for this simplified analysis, we assume their results to be trustable.

officially defined feed-in tariff for tidal current turbines in France, we use the wind energy tariff as a proxy. According to Tazi and Bouzidi (2019), France has implemented various policies to support the development of wind energy. A key measure introduced in 2000 is the purchase obligation agreement established by the French state. It guarantees a fixed feed-in tariff for wind-generated electricity, higher than the market electricity price, under a 10-year contract. For the first 10 years, the tariff is set at 8.2 ce/kWh. Due to uncertainty in the electricity price estimation, a Monte Carlo simulation was performed to obtain the empirical distribution of the price when a parameter is uncertain. This approach generates a distribution of potential electricity prices along with their expected values and allows the computations of the Net Present Value (NPV) under different electricity price scenarios.

According to the paper from Marjan and Shafiee (2018) the authors used a normal distribution for their Monte Carlo simulation for the spot price of electricity.

Therefore, here, the Monte Carlo simulation relies on a normal distribution too for the price of electricity resale of Normandie Hydroliennes, with the following parameters : a mean equals to 0.082 (which corresponds to the fixed feed-in tariff for wind-generated electricity of 0.082 euros/kWh) and a standard errors of 0.079 obtained by taking the monthly wholesale electricity price in France for 10 years. We use the wind-electricity feed-in tariff as we do not have value for the tidal current turbines electricity. This assumption is justified by the structure of the French energy mix. France relies heavily on nuclear power, which is largely depreciated and also provides low-carbon electricity at relatively low cost. Consequently, the policy incentive to offer high feed-in tariffs for alternative renewable sources such as tidal energy is weaker than in countries like the UK per se. Hence, the wind tariff provides a conservative yet plausible benchmark for the NH1 project's expected revenue. The simulation is composed of 20,000 iterations and we assume that electricity price and OPEX are uncorrelated.

6.1.2 CO_2 reduction

One of the most important externality for this project is the CO_2 emission that generates each unit of electricity produced. We do not have a precise value for those particular tidal turbines but Walker et al. (2015) find a value between 18 and 35gCO_2/kWh. Therefore, we build 3 different scenarios for this energy, a pessimistic one with a value of 35gCO_2/kWh, a realistic one with a value of 26.5gCO_2/kWh and an optimistic one with a value of 18gCO_2/kWh. Importantly, the analysis must be made relatively to a counterfactual. Hence, we use the French energy-mix value which acccording to Statista [4] is 56gCO_2/kWh. This allow us to get a difference in the CO_2 emission between the French energy-mix and the NH1 project (ΔCO_2). For simplicity, we take those values as fixed for the next 25 years. Recall, the pilot farm produce 33.9GWh which is the equivalent of 3.39e+7kWh. Using those values we can easily derive the

[4] Quelle est l'empreinte carbone de l'électricité produite en Europe ? (13 septembre 2024)

total CO_2 reduction per year for the realistic scenario:

$$\begin{aligned}
\Delta CO_2 \,/\, kWh &= (56 - 35)\,\text{g } CO_2/\,\text{kWh} = 21\,\text{g } CO_2/\,\text{kWh} \\
\Delta CO_2 \,/\, \text{year} &= 21\,\text{g } CO_2/\,\text{kWh} \times 3.39 \times 10^7\,\text{kWh/year} \\
&= 7.12 \times 10^8\,\text{g } CO_2\,/\text{year} \\
\Delta CO_2 \,/\, \text{year} &= 712\,\text{t } CO_2/\,\text{year}
\end{aligned}$$

6.1.3 Employment

We do not include gross wages as a separate benefit stream because these constitute transfer payments already recorded in CAPEX/OPEX. Instead, we adjust the labour component of costs by a shadow-wage factor to capture the net social opportunity cost of labour as recommend by Pearce et al. (2006). Indeed, they argue that in a context where there is no full-employment, the project or policy creates employment for labour that would otherwise be unemployed. This treatment avoids double-counting and isolates the true welfare effect of employment creation. While the literature does not identify a universally accepted labour share of CAPEX, International Renewable Energy Agency (IRENA) (2021) find that for 25 hydropower projects, the weighted average of the salary represents 39% of the OPEX. For the CAPEX, we proceed as follow: we take the 400 jobs annouced by NH1, we multiply them by the average salary in the energy, water and waste sector (2869€)[5], then we get how much it represents of the 90M of CAPEX[6]. To this part of labor, we apply the shadow wage factor of 85% as according to Conseil d'orientation pour l'emploi (2007), at the minimum wage level, a job seeker receives 85% of their former net salary. We obtain the following factors:

$$\text{Social Net CAPEX} = \text{CAPEX} \times (0.7 + \mathbf{0.3 \times 0.85}) - \text{European Subsidies} \tag{1}$$

$$\text{Social Net CAPEX} = 0.955 \times \text{CAPEX}$$

$$\text{Social Net OPEX} = \text{OPEX} \times (0.61 + \mathbf{0.39 \times 0.85}) \tag{2}$$

$$\text{Social Net OPEX} = 0.9415 \times \text{OPEX}$$

6.2 Costs

6.2.1 Capital Expenditures

Regarding the costs of the NH1 project, the main expense to consider is the capital expenditure (CAPEX), which amounts to €90 million at $t = 0$. Etim et al. (2024) defines the capital expenditure (CAPEX) as *"the funds a company invests in acquiring, maintaining, or upgrading physical assets to improve or expand its operations, generating future benefits, or extending the useful life of existing assets"*.

[5] INSEE, La France et ses territoires, Éditions 2021.

[6] For simplicity, this assumed no employer social contribution, that is, the mandatory payments made by firms in addition to gross wage.

6.2.2 Estimation of Operating Expenditures

The operating expenditures called (OPEX) is defined by Council and Finance (2013) as *"the total annual operating expenditure from the first year of a project's operation, given in per unit of installed capacity terms"*. Musial et al. (2021) highlights that operating expenditures (OPEX) are more significant for offshore wind energy than for onshore wind. Indeed, wind and wave conditions can limit maintenance personnel access to the sites, leading to additional costs such as vessels or helicopters. OPEX for offshore wind turbines is defined to be in a range between 70–80 $/kW (cost per kilowatt installed) expressed in real terms. Furthermore, industry analysts estimate that OPEX will decrease by 14–29% from 2020 to 2030 due to technological advancements.

For this analysis, we use the mean of 14–29%, corresponding to a 21.5% reduction in OPEX over 10 years. Annualized, this gives a 2.39% reduction per year, which will be applied over 25 years as follows:

$$(1+r)^{10} = 1 - 0.215 \quad \Rightarrow \quad r \approx -2.39\% \tag{3}$$

where r represents the annual reduction rate of OPEX caused by technical progress.

7 Step 6 : Monetize consequences

Several consequences considered in this analysis are already quantified in euros value and in real terms. Thus, we do not require further modification. It concerns the following consequences : Social Net CAPEX, Social Net OPEX and Electricity Revenues.

However, we still need to monetize the CO_2 emission reduction resulting from the NH1 project. To address this, we use the *valeur de l'action pour le climat* **(VAC)**[7] published by *France Stratégie (2025)*[8] . The value represents the social cost of carbon for France and serves as official in public climate-related investment evaluations. That corresponds to the minimum economic value the society should assign to emissions reduction, in order to stay in the pathway of the *Stratégie national bas-carbone* (SNBC)[9]. France Stratégie (2025) sets the value (in 2023€) at 256€/tCO_2e in 2025, 300€/tCO_2e in 2030 and 563€/tCO_2e in 2050.

8 Step 7 : Net present value of costs and benefits

Computing the net present value (NPV) of the NH1 project is the final stage of the analysis. This allows us to determine whether the benefits outweigh the costs and to conclude whether the project is socially efficient.

[7]Translation : The climate action value.

[8]France Stratégie (2025), *La valeur de l'action pour le climat. Une référence pour évaluer et agir*, rapport de la commission présidée par Alain Quinet, mars 2025.

[9]Translation : Low-carbone national strategy.

As mentioned earlier, uncertainty regarding the electricity resale price results in multiple NPV values, depending on the different price levels generated by the Monte Carlo simulation. The net present value (NPV) is computed as follows:

$$\text{NPV}^{(i)} = \left(\sum_{t=0}^{25} \frac{B_t^{(i)}}{(1+r)^t} \right) - \left(\sum_{t=0}^{25} \frac{C_t^{(i)}}{(1+r)^t} \right), \quad i = 1, 2, \ldots, 20{,}000 \tag{4}$$

$B_t^{(i)}$ = Benefits include electricity revenues, CO_2 reduction and employment creation in real terms
$C_t^{(i)}$ = Costs include CAPEX and OPEX in real terms
$T = 25$ = project lifetime (years)

In CBA, it is assumed that individuals' preferences can be aggregated so that social benefit is the sum of all individuals' benefits and social cost is the sum of all individuals' costs (OECD, 2018).

One last concern is the discount rate. Gollier (2012) highlights that, to obtain an optimal decision rule in CBA, the discount rate should correspond to the rate of return on risk-free investments in the economy. Gollier therefore recommends using a discount rate of around 2.5–3% for risk-free projects with a 20-year horizon, rather than 4%. In our case, the applied discount rate is 2.75%, which is smaller than the upper bound of 3% as we are operating with a slightly longer time horizon of 25-year.

After clearly defining the setting, we present the evolution of the NPV (in million euros) over time under three scenarios and for a fixed electricity price of the average value (0.082 euros/kWh), as described earlier: a pessimistic scenario with 35gCO_2/kWh, a realistic scenario with 26.5gCO_2/kWh, and an optimistic scenario with 18gCO_2/kWh, shown in Figure 2. We see that it is strongly increasing overtime, which could be explain by a large positive difference between marginal benefits and marginal costs. Hence, the bigger burden is on the fixed investment at t=0 as it appears in Figure 1. Additionally, Figure 3 represents the empirical distribution of the NPV for the different scenarios. Considering the realistic scenario, we get :

$$P(\text{NPV} < 0) = 77\%$$

In other words, under the realistic scenario, there is a probability of 77% of obtaining a negative NPV. Therefore, it can be inferred that under this probability, the project NH1 is not socially efficient in a CBA framework.

9 Conclusion

This study conducts a cost-benefit analysis (CBA) to determine whether the NH1 project is socially efficient and should therefore be implemented. The NH1 project is France's first pilot tidal turbine farm, located in the Raz Blanchard area, which is characterised by exceptionally strong currents. The project will contribute to diversifying the French energy mix, reducing carbon emissions, and providing a French local village with a full energy supply. From a decision-making perspective, the CBA is conducted over a 25-year horizon, which corresponds to the expected lifetime of the four installed turbines. The benefits are considered to be the revenue generated, the reduction in CO_2 emissions and employment, whereas the costs include CAPEX and OPEX expenditures. One limitation of this analysis is the uncertainty surrounding the resale electricity price and the CO_2 emissions of the four turbines. Therefore, the analysis incorporates this uncertainty through a Monte Carlo simulation and three different scenarios. Considering the realistic scenario with 26.5gCO_2/kWh emissions, there is a 77% probability of a negative NPV [10]. Thus, based on this result, under the current assumptions, the NH1 project does not appear to be socially efficient, as the costs outweight the benefits generated. The results also indicate that the NPV is highly sensitive to electricity prices, which are inherently uncertain. In addition, another limitation of the analysis is that it takes only into account a 25-year time horizon, while the project expect to expand its tidal power in Normandy in a longer-time horizon with new technologies and cost reductions through R&D.

Overall, although the current CBA indicates that the NH1 project is not socially efficient, its long-term benefits such as the accumulation of technological experience, potential future cost reductions, and opportunities for expanding tidal energy capacity should be taken into account. The results should be interpreted cautiously, given the NPV's strong sensitivity to electricity prices and the limitation of a 25-year analysis horizon. The project may hold a great potential for contributing on the leaning-by-doing process and reducing future costs in the tidal energy sector.

[10] For the optimistic scenario the probability is of 70.46%, and for the pessimistic one, the probability is of 83.71%.

10 Appendix

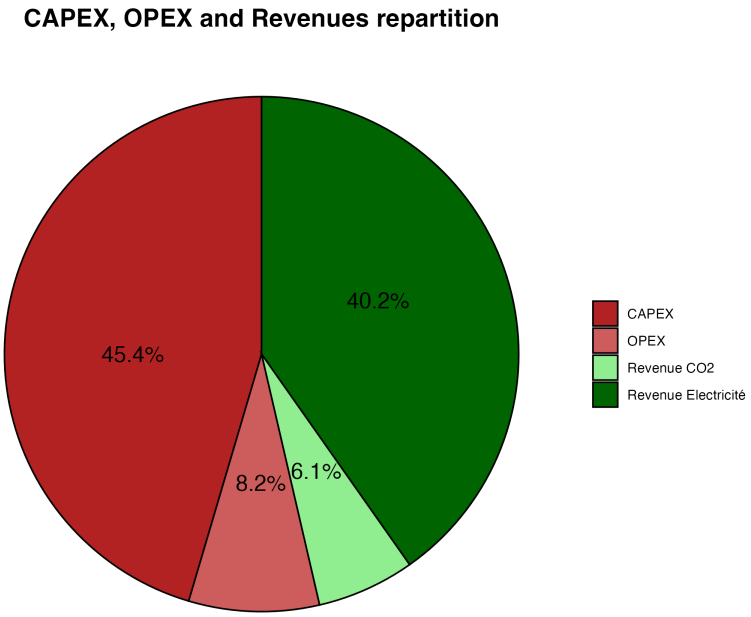


Figure 1: Contribution of each benefits and costs to the NPV (in absolute value).

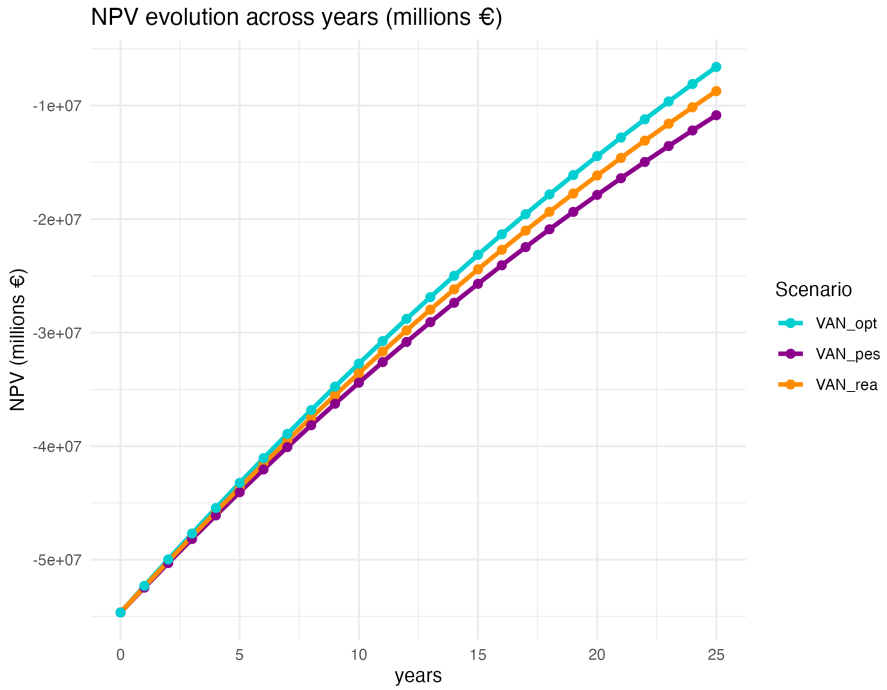


Figure 2: Evolution of the NPV across years and scenarios for a fixed electricty price.

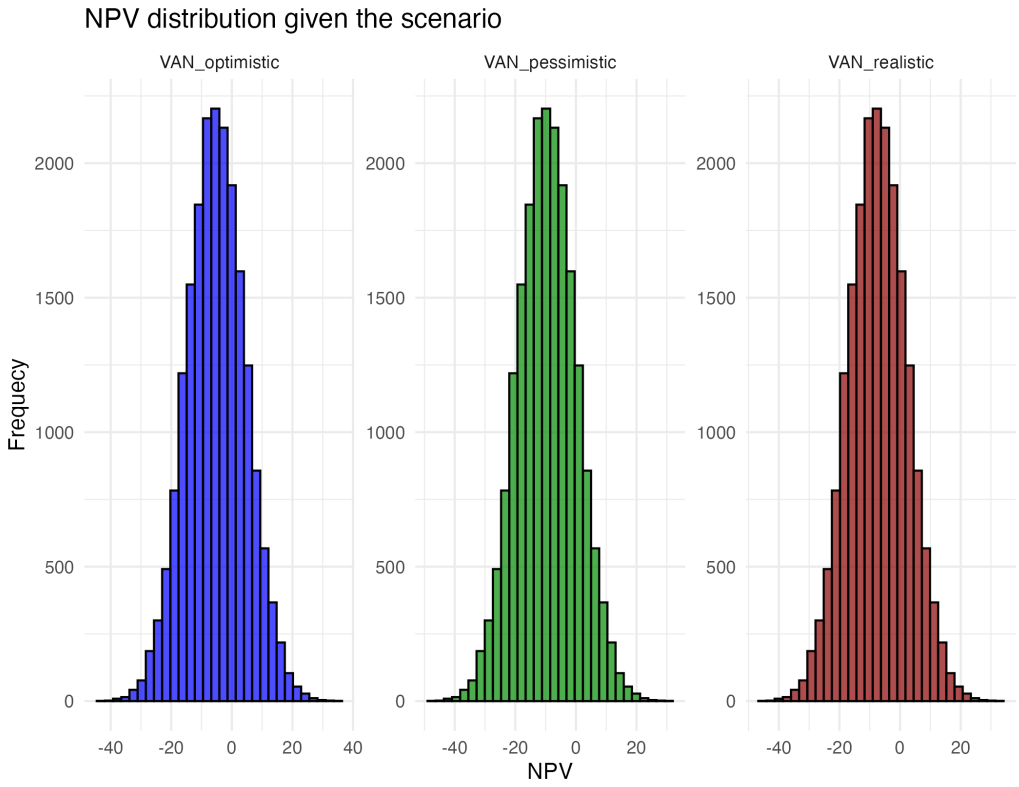


Figure 3: NPV distribution across years and scenarios.

11 AI Acknowledgment

We made acknowledge having made use of AI (ChatGPT) during this project. Specifically:

- We have sent some documents and ask whether specific information were inside them (e.g. Does this paper provide an estimation for electricity price). Then, it allows us to search directly in the document.
- We used AI to help reorganize some paragraphs by putting our own points in a clearer order. For example, we send how each consequence was quantified and why. Then we asked how would you structure that.
- Providing the AI with the parameters and distribution we want to use, we asked for code's sketch on how to implement a Monte Carlo analysis.

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